

the training process while the ANTI-CURRICULUM approach would lead to the opposite. According to the recent debate on the correlation between sharpness and generalization ability of foundation models [Dinh et al., 2017, Foret et al., 2021, Andriushchenko et al., 2023], it is fascinating to uncover whether the curriculum learning algorithm could realize similar outcome as computation-heavy sharpness-aware minimization optimization.

7 Conclusion

In this paper, we propose IRREDUCIBLE CURRICULUM as a curriculum learning algorithm with the underlying hypothesis that small proxy model is capable to simulate the early stage training trajectory. For language model pretraining, the method shows a consistent improvement in validation perplexity across several domains and downstream few-shot inference accuracy on the MMLU benchmark. Besides, we also observe our IRREDUCIBLE CURRICULUM can lead to a lower sharpness compared to the random baseline. However, because of the limitation of computation resources, we leave the scaling-up experiments and exploration of further generalization patterns to future investigations.

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A Preliminary of Hessian and Sharpness

Hessian. Considering model parameter θ and input x_i , the hessian is defined as the average second-order derivative of the objective function ℓ :

$$\nabla_{\theta}^2 \mathcal{L}(f(\theta)) = \frac{1}{n} \sum_{i=1}^n \nabla_{\theta}^2 \ell(f(\theta, \mathbf{x}_i)) \quad (5)$$

Specifically, we randomly sample 50 instances from validation set of each domain, and take the average second-order derivative as the hessian matrix. We only consider parameters from all dense layers as θ while skip the attention matrix.

Sharpness. Denote the hessian matrix as $\mathcal{H}$, we adopt a popular definition of the sharpness as the largest eigenvalue of the Hessian $\lambda_{\max}(\mathcal{H})$, following [Wu, 2018, Li et al., 2022b].

When calculating the sharpness, we apply the numerical method, power method, following [Golmant et al., 2018] to calculate the top eigenvalue of the Hessian matrix.

B Related Work

Curriculum Learning for Language Modeling. As a popular concept in education and cognitive science, there have been plenty of attempts to apply curriculum learning on training language models, by progressively expose the neural learner with samples with ascending difficulty level [Campos, 2021, Nagatsuka et al., 2021, Li et al., 2022a]. However, most of the effective methods increase the difficulty by varying the context length of sequences, without focusing on the syntactic nor semantic properties. Li et al. [2022a] empirically proves that with a incrementally increase context length, the pretraining of language model could be more stable, reducing the number of loss spikes. Similarly, Nagatsuka et al. [2021] shows that feed training samples with increasing block size could improve the performance when pretraining BERT [Devlin et al., 2019], an encoder-only model with masked language modeling objective.

Campos [2021] has conducted extensive experiments towards using linguistic-featured curriculum learning for language model pretraining. However, the conclusion shows their curriculum fails to effectively help with validation perplexity nor downstream performance.

Recently, Xie et al. [2023] has introduced a domain reweighting framework by sampling from domains with various probability. They propose to use two small-scale auxiliary models to determine the domain weights through a group distributional robust optimization game. However, all the samples within one domain would share the same sampling weights, which fails to distinguish the instance-wise attributes inside the domain barrier. How to combine the domain-level reweighting and the in-domain curriculum could also be a promising direction for future exploration.

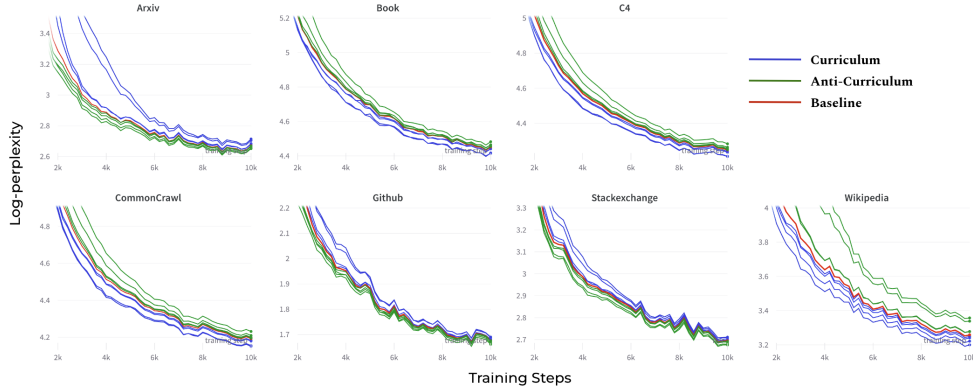


Figure 5: Domain-wise validation log-perplexity on RedPajama-1B.